Satwik Paladugula

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GitHub — LinkedIn

Profile

Computer Science undergraduate with hands-on experience in backend engineering, data analysis, and AI application development. Demonstrated expertise in Python, SQL, Java, FastAPI, and database design with a proven track record of architecting secure reverse proxies, automated payment engines, and natural language database systems. Certified in ServiceNow (CAD/CSA) and SAP ABAP Cloud.

Education

B.Tech — Computer Science and Technology **2023–2027**
Sri Vasavi Engineering College, Tadepalligudem **CGPA: 8.25/10**
Intermediate (PCM) **2021–2023**
Bharatiya Vidya Bhavans 84.8%
SSC **2020–2021**
Tirumala Proactive School **10/10 CGPA**

Technical Skills

- **Programming & Languages:** Python, SQL, Java, C/C++
- **Frameworks & Backend:** FastAPI, Spring Boot, Spring Data JPA, Hibernate, Pydantic, RESTful APIs, Starlette
- **AI & Security Engineering:** LLM Integration (Gemini), Prompt Injection Defense, SQLGlot, Sensitive Data Masking, Shannon Entropy
- **Data Analysis & Storage:** MySQL, SQLite, DBMS, Pandas, NumPy, Matplotlib, Data Validation & Auditing
- **FinTech & Tools:** NPCI UPI Integration, Dynamic QR Generation, ReportLab Engine, Git, GitHub, VS Code, Postman
- **Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Computer Networks

Projects

AI Security Gateway — Enterprise LLM Proxy & Prompt Defense Python, FastAPI, Streamlit, SQLite, Pydantic

- Engineered a production-grade reverse proxy and policy enforcement point between enterprise clients and LLMs to prevent confidential data leakage and mitigate prompt injection attacks.
- Implemented pluggable sensitive data detectors with algorithmic validation (Luhn checksum for credit cards, ISO 7064 Mod 97-10 for IBANs, and Shannon entropy for API keys), achieving sub-millisecond added latency (0.14 ms p50).
- Developed stateless deterministic multi-turn masking (<GW_ENTITY_N>) with safe response unmasking, layered heuristic jailbreak detection, and privacy-preserving SQLite audit logging using salted SHA-256 hashes.
- Designed an interactive Streamlit operations console with side-by-side prompt inspection, verified across a 550-prompt synthetic evaluation benchmark with 90% unit test coverage.

CollectFast — 0% Fee UPI Invoicing & WhatsApp Recovery Engine Python, FastAPI, ReportLab, SQLite, UPI Protocol

- Architected an automated B2B invoicing SaaS enabling businesses to create GST tax invoices and collect direct payments with 0% gateway transaction fees via native NPCI UPI protocol.
- Implemented dynamic real-time UPI QR generation and mobile deep-linking triggers (upi://pay?pa=...) enabling instant 1-tap checkout via Google Pay, PhonePe, and Paytm.
- Engineered a 1-click WhatsApp payment recovery engine with automated behavioral reminder templates (Friendly, Due Date, Overdue), accelerating client payment recovery times by 70%.
- Built an automated server-side PDF generator using ReportLab with embedded high-density payment QR codes, itemized tax computations, and automated client ledger balances.

ClarifySQL — AI-Powered Natural Language Data Assistant Python, FastAPI, Gemini, MySQL, SQLGlot, Streamlit

- Developed an AI-powered data assistant that converts natural language business questions into secure, schema-aware SQL queries for MySQL databases.
- Implemented query understanding, ambiguity detection, interactive clarification workflows, and strict AST-level SQL validation using Python and SQLGlot to prevent unauthorized mutations.
- Built a FastAPI REST backend and responsive Streamlit interface featuring dynamic query result visualization, persistent execution history, and session-based conversational context.

Certifications

- SAP Certified - Back-End Developer - ABAP Cloud
- ServiceNow Certified System Administrator (CSA)
- ServiceNow Certified Application Developer (CAD)
- Oracle Generative AI Professional
- Salesforce Agentforce Specialist

Strengths

- **Analytical & Logical Problem Solving:** Strong algorithmic thinking, debugging, and systems breakdown skills
- **Autonomous & Rapid Learner:** Proven ability to master enterprise frameworks, cloud technologies, and emerging AI tools
- **Technical Communication:** Clear and structured verbal and written technical documentation and API specifications
- **Teamwork & Collaboration:** Effective cross-functional collaboration in agile engineering and development environments